

ORIGINAL ARTICLE

Aphrodisiac properties of hydro-alcoholic extract of *Cassia auriculata* flower in male rats

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Abstract

Cassia auriculata is a commonly found plant in Asia, widely used in Ayurveda and Siddha medicines as a tonic, astringent and in general for diabetes. Herbal tea made from this plant has been marketed as a product for restoring sexual vitality, to increase sperm count and counteract ejaculatory disorders. However, the scientific evidences are scarce to prove this concept. Here, we examined the effect of hydro-alcoholic extract obtained from *C. auriculata* flower upon the expression of male Wistar albino rat's sexual behaviour. Sildenafil was used as a positive control. Penile erection index (PEI), mount latency (ML), intromission latency (IL), ejaculation latency (EL), mounting frequency (MF), intromission frequency (IF), ejaculation frequency (EF) and post-ejaculatory interval (PEI) were recorded for days 0, 7, 14 and 28 and also after the withdrawal of the treatment on days 7 and 15. Significant reduction in ML, IL and PEI, and increment in EL, PEI, MF, IF and EF were observed ($p < 0.05$, < 0.01). However, neither extract nor sildenafil sustains the effect after withdrawal of the treatment. The present finding demonstrates the aphrodisiac potential of hydro-alcoholic extract of *C. auriculata* flower *in vivo* and lends support to the traditional utilisation as a sexual stimulating agent.

KEYWORDS

animal model, *Cassia auriculata*, erectile dysfunction, PDE5

1 | INTRODUCTION

A drug or formulation of food capable of inducing sexual desire, increases sexual pleasure and performance could be considered as aphrodisiac. The word "Aphrodisiac" is derived from "Aphrodite" the Greek goddess of love (Greek-Aphrodisiakos-sexual) (Singh, & Mukherjee, 1988). Sandroni (2001) has classified the aphrodisiacs by their mode of action into three types: (a) substances that increase libido: these aphrodisiacs act at the level of the central nervous system by altering specific neurotransmitter or specific sex hormone concentrations particularly testosterone concentration, for example, ambrein from *Ambra grisea* (Taha, Islam, & Ageel, 1995); (b) substances that increase potency (allow or sustain erection): these

compounds generally act through induction of vasodilation, for example, sildenafil citrate an oral drug for men with erectile dysfunction, produce satisfactory erections and improve sexual satisfaction without affecting sexual desire (Jackson, 1998); and (c) substances that enhance the sensory experience during coitus: the effect of such substances is generally through irritation of the genital mucosa; for example, cantharidin is a chemical with vesicant properties derived from blister beetles, and has been used for millennia as a sexual stimulant. In general, aphrodisiacs are mainly used to treat the male sexual dysfunction, that is, to increase libido, potency or sexual pleasure (Karras, Farrell, Harrigan, Henretig, and Gealt, (1996). Many plants with the medicinal properties are reportedly effective as aphrodisiacs either to increase libido or potency; for example, extracts from

Lepidium meyenii Walp. (maca), *Panax ginseng* C.A. Meyer, *Ginkgo biloba* L., *Ferula hermonis* Boiss. and many other herbal remedies, alone or in combination, have been promoted for the treatment of sexual problems. A variety of natural plant products, including berberine, forskolin, papaverine, yohimbine and icariin, are claimed to be useful for improving sexual performance (Dell'Agli, Galli, Vrhovsek, Mattivi, & Bosisio, 2005; Melnyk, & Marccone, 2011). Ethnobotanical surveys have also indicated a large number of plants traditionally used as aphrodisiacs, but only few of them are scientifically authenticated for the management and treatment of the male sexual dysfunction (Malviya, Malviya, Jain, & Vyas, 2016). The allopathic drugs used for the sexual dysfunction are believed to produce a variety of adverse effects and ultimately affect the general health. Therefore, the search for natural supplement from medicinal plants is being intensified probably because of less side effects as well as higher availability and affordability.

Cassia auriculata L. commonly known as tanner's cassia, also known as "avaram" in Tamil language and Ranawara in Sri Lanka, a shrub belongs to the Caesalpiniaceae family. The shrub is especially famous for its attractive yellow flowers, which are used in the treatment of skin disorders and body odour. It is widely used in traditional medicine for rheumatism, conjunctivitis, diabetes and as tonic (Joshi, 2000). It is one of the principle constituents of "Avaarai panchaga chooranam"—an Indian herbal formulation used in the treatment of diabetes to control the blood sugar level (Brahmachari & Augusti, 1961). Compound syrup prepared using the *C. auriculata* flowers, mocharas and Indian saparilla are proposed for nocturnal emissions. The flowers of the plant are used in preparation of tea, suggested for diabetes. The seeds are used in diabetes, opthalmia and chylous urine (Kirthkar & Basu, 1965). Ranawara flower has been widely used from centuries in Sri Lanka as a caffeine-free Herbal tea. Ranawara herb tea is also widely available in the market as a product in restoring the sexual vitality, increases the sperm count and counteracts the ejaculatory disorders. Diabetic is believed to be managed by the tea prepared for using dried flower and buds of *C. auriculata*. It is also supposed to improve the complexion in women (Jayaweera, 1981). Its bark is used as an astringent, leaves and flowers as anthelmintic, seeds used to treat in eye troubles, roots employed in skin diseases, and the leaves also have laxative properties (Siva, & Krishnamurthy, 2005).

Previous literature studies on crude extracts of this plant have documented the scientific basis for some of its traditional medicinal uses. Accordingly, the roots extract of *C. auriculata* has been shown to possess nephroprotective activity in cisplatin and gentamicin-induced renal injury (Annie, Rajagopal, & Malini, 2005). Additionally, the leaves extract has shown to possess antihyperglycemic (Gupta, Sharma, Bansal, & Prabhu, 2009; Sabu, & Subburaju, 2002), hypolipidemic (Gupta et al., 2009), hepatoprotective activity from alcohol-induced injury (Kumar, Ponmozhi, Viswanathan, & Nalini, 2003), antipyretic (Vedavathy, & Rao, 1991) and *in vitro* anticancer effects (Prasanna, Harish, Pichai, Sakthisekaran, & Gunasekaran, 2009). The antipyretic effect of alcoholic extract of *C. auriculata* has been proved (Vedavathy, & Rao, 1991); also, hepatoprotective (Chauhan, Patel, Valera, Patil, & Surana, 2009) and anti-diabetic (Pari, & Latha,

2002; Surana, Gokhale, Jadhav, Sawant, & Wadekar, 2008) activities have documented in the literature. Manimegalai and Venkatalakshmi (2012) reported the cardioprotective effect of *C. auriculata* petal extract on isoproterenol-induced myocardial infarction in albino rats. The methanolic extract of the flower also possesses the strongest antiviral activity against herpes simplex-1 (Arthanari et al., 2013). Thus, the medicinal properties of *C. auriculata* have been vastly proved and accepted. Previously, we have investigated the *in vitro* antioxidant activity of *C. auriculata* flower extract (Kumaran, & Karunakaran, 2007) and presently reporting the aphrodisiac properties. To the best of our knowledge, no modern scientific literature is available for the aphrodisiac activity of *C. auriculata* flower.

2 | MATERIALS AND METHODS

2.1 | Plant collection and preparation of hydro-alcoholic extract

Cassia auriculata flower was collected from the area in Aruppukottai, Tamilnadu, India. A voucher specimen (HLL/RD/CA14A) identified by a reputed botanist, Dr. P. Santhan, M/s. Durva Herbal Centre, Chennai, India, and was deposited in our institute for future reference. The flowers were dried at a hot air oven with air circulating fan (Rotak Instruments, Kerala, India) (35–40°C) and were ground to powder and passed through a sieve (60 mesh). About 100 g of the powdered flower was extracted with 300 ml of 50% ethanol in water at 60°C for 3 hr on a water bath and then filtered. The extraction process was repeated two more times, with fresh solvent each time, and the combined filtrate was concentrated under vacuum in a rotary evaporator (400 mm of Hg) at 60°C and obtained a dry hydro-alcoholic extract (20.7 w/w %) of *C. auriculata* flower (CAE).

2.2 | Animals

Inbred, sexually active Wistar albino rats (24 male and 24 female), weighing about 160–190 g (11–13 weeks old), were used for the sexual behaviour study. The male rats were trained for the sexual behaviour, that is, two times a day for 10 days. The male rat which did not show any sexual interest during the test period was considered as an inactive male. The sexually active male rats were selected for testing aphrodisiac activity of the extracts. The female rats were brought in oestrous phase artificially by sequential administration of estradiol benzoate (Sigma-Aldrich) (25 µg/rat) and progesterone (Sigma-Aldrich) (250 µg/rat) subcutaneously, 48 hr and 4 hr prior to experimentation, respectively. The drugs were dissolved in arachis oil (Sigma-Aldrich) and injected subcutaneously in a volume of 0.1 ml/rat. The receptivity of the female animals was confirmed before the test by exposing them to male animals, other than the control and test animals. The most stimulus receptive females were then selected for the study (Saad, Eid, Abd El-Latif, & Sayed, 2013). Any female animal that does not show receptivity was replaced with another artificially "warmed" female. For toxicity study, 12 healthy male albino rats were used separately. All animal experiments

were performed as per the protocols and recommendation of the Institutional Animal Ethics Committee of HLL Lifecare Limited, Thiruvananthapuram, India (Reg no. 1782/PO/Rc/S/14/CPCSEA) and were in accordance with the international standard on the care and use of experimental animals. The rats of either sex were isolated and housed in separate cages at room temperature ($24 \pm 2^\circ\text{C}$) on a reversed day-night cycle (06.00:18.00) with relative humidity of 50%–55% during the course of the experimental period. The animals were fed with a standard pellet diet and provided water ad libitum.

2.3 | Grouping and dosing of animals

The male rats were randomly divided into four groups of six animals each. Group I served as a control (CL) and were given vehicle, 1% sodium carboxymethyl cellulose (CMC) in water. The Group II and Group III were treated with CAE at two different doses of 100 mg/kg (CAE 100) and 200 mg/kg (CAE 200), respectively. The IV group was treated with sildenafil citrate (SC) at a dose of 5 mg/kg (Kenmogne et al., 2016). The extract and sildenafil citrate were prepared in an aqueous solution containing 1% sodium CMC. All rats were treated orally in a once-daily regimen for a period of 28 days.

2.4 | Toxicity study

Acute toxicity studies were carried out as per the guidelines (no. 423) given by the Organization for Economic Co-operation and Development (OECD) (Acute oral toxicity test method, 2001). Overnight fasted healthy male albino rats were used for this study (grouped into four and each group consisted of three rats). The extracts were administered to all the three animals in the first group at a starting single dose of 5 mg/kg. Animals were observed for a period of 2 hr, then occasionally for 4 hr for severity of any toxic signs and mortality. Since no mortality was observed at this dose, the same procedure was repeated for dose levels of 50, 300 and 2,000 mg/kg of extracts on separate newer groups. The behavioural changes closely observed for were as follows: hyperactivity, ataxia, tremors, convulsions, salivation, diarrhoea, lethargy, sleep and coma. The following general behaviours were also observed for any abnormalities viz-a-viz locomotion, wall climbing, rearing, grooming, scratching, eating, drinking and digging. The animals were kept under observation up to 14 days after drug administration to find out any delayed mortality (Sharma, Thakur, & Dixit, 2012).

2.5 | Functional studies

2.5.1 | Penile erection index (PEI)

Penile erection was determined by using the method reported by Thakur et al. (2011). In brief, each male rat was placed in a transparent observation cage ($60 \times 40 \times 40$ cm) that was divided into half by 2 sheets of plastic fibre mesh, preventing contact but allowing auditory, visual and olfactory stimuli. After a 5-min adaptation period, the test was started by placing an oestrus female on the other side

of the cage. The numbers of erections were recorded. Erection in rats was marked by the visibility of the penis out of its sheath or by grooming of the penis, which is another indicator of penile erection in rats. Penile erection index (PEI) was calculated as per the methodology reported by Islam, Tariq, Ageel, Al-said, and Al-Yhya (1991), that is, by multiplying the percentage of rats exhibiting at least one episode of penile erection during 30-min observation period with the mean number of penile erections: $\text{PEI} = \text{percentage of rats exhibiting penile erection} \times \text{mean number of erections}$.

2.5.2 | Copulatory behaviour testing

The mating behaviour tests were carried out using the method reported by Dewsbury and Davis (1970). Sexual behaviour in male rats consists of three distinct phases: (a) Mount: where the male rat assumes the copulatory position, but does not insert its copulatory organ (the penis) into the vagina; (b) Intromission: the copulatory organ enters the vagina during a mount; and (c) Ejaculation: forceful expulsion of semen. Mating behavioural testing was performed under dim red lights. The mating test was performed 2 hr after the beginning of the dark phase (Agmo, 2003). The test was started by placing males in the observation cage ($60 \times 40 \times 40$ cm), and then, after 10 min, a stimulus receptive female was presented (Almeida, Kempinas, & Lamano-Carvalho, 2000). Upon presentation of female, the tests lasted for 30 min and the following parameters were recorded according to the method of Butcher, Butcher, and Larsson (1969): (a) Mount frequency (MF): the number of mounts without intromission from the time of introduction of the female until ejaculation; (b) Intromission frequency (IF): the number of intromissions from the time of introduction of the female until ejaculation; (c) Mount latency (ML): the time interval between the introduction of the female and the first mount by the male; (d) Intromission latency (IL): the time interval from the time of introduction of the female to the first intromission by the male; (e) Ejaculation latency (EL): the time interval between the first intromission and ejaculation (characterised by longer, deeper pelvic thrusting and slow dismount followed by a period of inactivity); (f) Ejaculation frequency (EF): the number of ejaculations in a session; (g) Post-ejaculatory interval (PEI): the time interval between ejaculation and the first intromission of a new series. All the copulatory events were recorded using a computerised night vision video camera recorder.

2.5.3 | Effects of drug after withdrawal of treatment

Daily dosing of the rats was stopped after 28 days of the treatment. The sexual behaviour of the animals was evaluated again on days 7 and 15 post-cessation of treatment (Sharma et al., 2012).

2.6 | Estimation of total phenolic content in CAE

The total phenolic content of the CAE was determined by the Folin-Ciocalteu method (Kaur, & Kapoor, 2002). Briefly, 200 μl of crude extract (1 mg/ml) was made up to 3 ml with methanol:water (1:1), mixed

thoroughly with 0.5 ml of Folin-Ciocalteu reagent for 3 min, followed by the addition of 2 ml of 20% (w/v) sodium carbonate. The mixture was allowed to stand for a further 60 min in the dark, and absorbance was measured at 650 nm. The total phenolic content was calculated from the calibration curve, and the results were expressed as mg of gallic acid equivalent per g dry weight.

2.7 | Estimation of total flavonoid content in CAE

The total flavonoid content of crude extract was determined by the aluminium chloride colorimetric method (Chang, Yang, Wen, & Chern, 2002). In brief, 50 μ l of crude extract (1 mg/ml methanol) was mixed with 4 ml of distilled water and 0.3 ml of 5% NaNO₂ solution. Then, 0.3 ml of 10% AlCl₃ solution was added after 5 min of incubation, and the mixture was allowed to stand for 6 min. Then, 2 ml of 1 molar NaOH solution was added, and the final volume of the mixture was brought to 10 ml with double-distilled water. The mixture was allowed to stand for 15 min, and absorbance was measured at 510 nm. The total flavonoid content was calculated from the calibration curve, and the result was expressed as mg rutin equivalent per g dry weight.

2.8 | Statistical analysis

All values are mean $\pm$ standard deviation (SD) and were compared with the control. The significance of the difference in the mean was determined by Student's *t* test. Values of $p < 0.05$ were considered as significant and $p < 0.01$ as highly significant. The data were analysed, and graphs were plotted with Excel Programme.

3 | RESULTS

3.1 | Toxicity study

The hydro-alcoholic extract was found to be safe up to 2000 mg/kg. No mortality or toxic symptoms were observed during the entire duration of the study, and also, no change in general behaviours such as locomotion, wall climbing, rearing, grooming, scratching, eating, drinking and digging was observed.

3.2 | Functional studies

3.2.1 | Effect of CAE on penile erection index

The administration of the extract resulted in a significant dose-dependent increase in PEI as compared to control group (Figure 1). The PEI was nearly two times higher on 15th day and three times higher on 28th day both in CAE dosed at 200 mg/kg body weight and in sildenafil citrate-treated group during the treatment period. Increased PEI in treated animals was indicative of the high degree of sexual stimulation. The PEI was got reduced after withdrawal of the treatment both in extract and sildenafil citrate groups.

3.2.2 | Copulatory behaviour testing

Effect of CAE on the latency of mating behaviour

The observations of latency of mating behaviour are presented in Figures 2–4. The ML and IL significantly decreased ($p < 0.01$), whereas the EL significantly increased ($p < 0.01$) at CAE 200 mg/kg dose as well as sildenafil citrate-treated group when compared to control. All these effects demonstrated that CAE stimulated elevated copulatory activities in the treated rats.

Effect of CAE on the frequency of mating behaviour

CAE increased the MF, IF and EF significantly ($p < 0.01$) at CAE 200 mg/kg dose-treated group when compared to the control on 15 and 28 days, which reflects an increase in sexual pleasure of extract-treated rats (Figures 5–7). The effects of sildenafil citrate in behavioural parameter could also be significantly observed on 15 and 28 days of this investigation.

Effect of CAE on post-ejaculatory interval

PEI reduced significantly in male rats and the value was less than half when compared with control group on 28th day in the CAE dosed at 200 mg/kg body weight and in the sildenafil citrate-treated group (Figure 8).

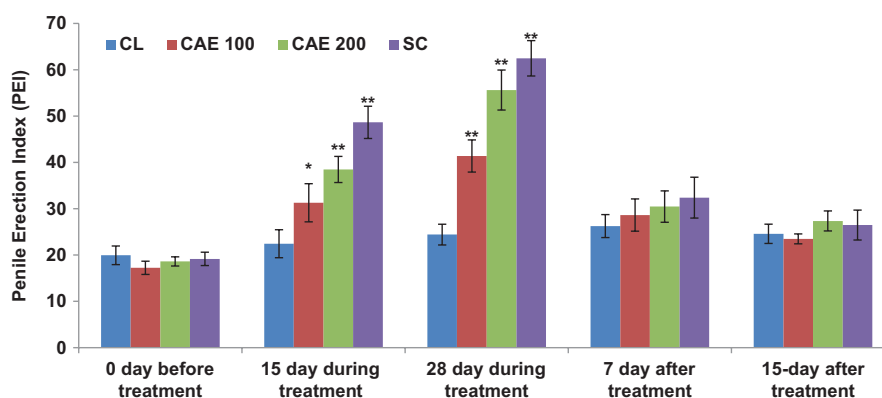


FIGURE 1 Effect of hydro-alcoholic extract of *Cassia auriculata* flower on penile erection index in male rats. Values are mean $\pm$ SD; $n = 6$; * $p < 0.05$, ** $p < 0.01$ vs. control

FIGURE 2 Effect of hydro-alcoholic extract of *Cassia auriculata* flower on mount latency in male rats. Values are mean $\pm$ SD; $n = 6$; * $p < 0.05$, ** $p < 0.01$ vs. control

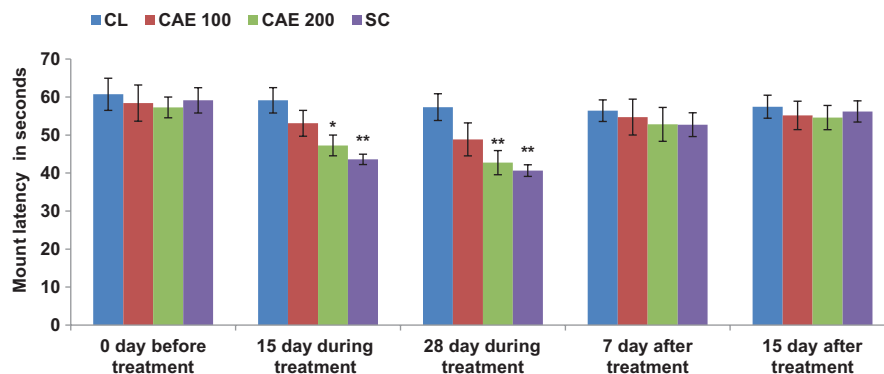


FIGURE 3 Effect of hydro-alcoholic extract of *Cassia auriculata* flower on intromission latency in male rats. Values are mean $\pm$ SD; $n = 6$; * $p < 0.05$, ** $p < 0.01$ vs. control

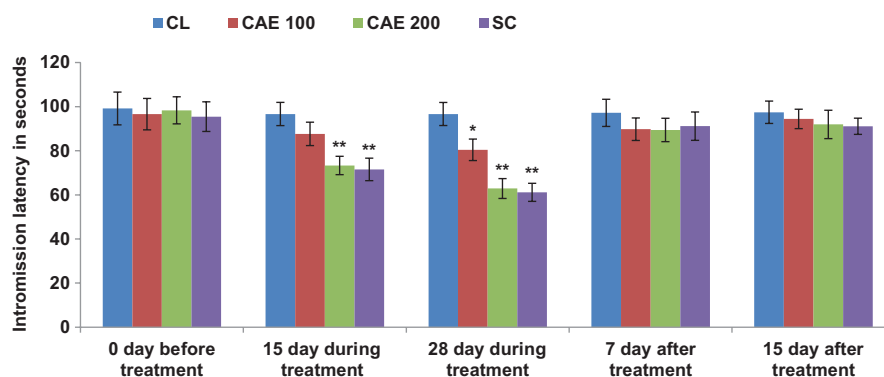
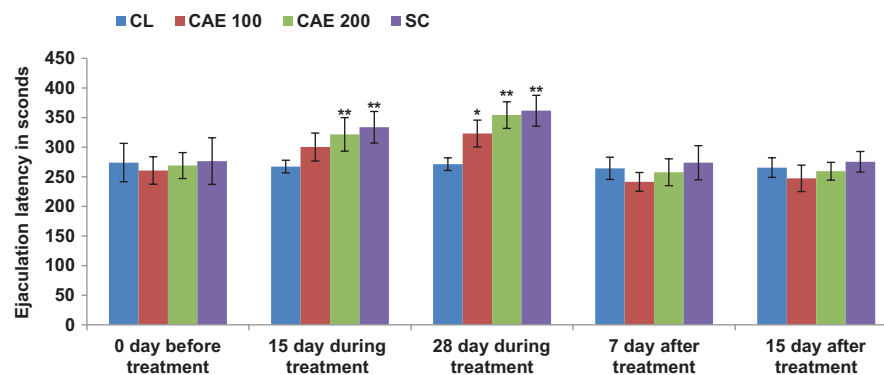


FIGURE 4 Effect of hydro-alcoholic extract of *Cassia auriculata* flower on ejaculation latency in male rats. Values are mean $\pm$ SD; $n = 6$; * $p < 0.05$, ** $p < 0.01$ vs. control



3.2.3 | Effects of drug after withdrawal of treatment

The copulatory behaviours of both CAE and sildenafil citrate treated animals, that is, latency, frequency of mating and PEI, were not changed significantly after withdrawing the treatment on 7 and 15 days, compared to the control group (Figures 2–8).

3.3 | Phenolic and flavonoid contents

The total phenolic content of the CAE, calculated from the calibration curve ($R^2 = 0.997$), was 141.07 ± 6.71 gallic acid equivalents/g, and the total flavonoid content ($R^2 = 0.998$) was 62.24 ± 3.27 rutin equivalents/g (Table 1).

4 | DISCUSSION

Plants have been used worldwide of varied cultures as a safe natural source of medicines for the treatment of sexual dysfunction (Kenmogne et al., 2016; Sharma et al., 2012; Tajuddin, Ahmad, Latif, & Qasmi, 2004), and hence, the need to screen more botanical resources of natural origin becomes inevitable. *C. auriculata*, a well-known herbal drug, is widely used in Ayurveda and Siddha for various ailments, particularly for diabetes and also available as herbal tea for sexual dysfunction, which has been chosen here to investigate the aphrodisiac activity. Before subjecting to sexual behaviour studies, we have investigated the toxicity of the hydro-alcoholic extract of flower of this plant. *C. auriculata* was used in

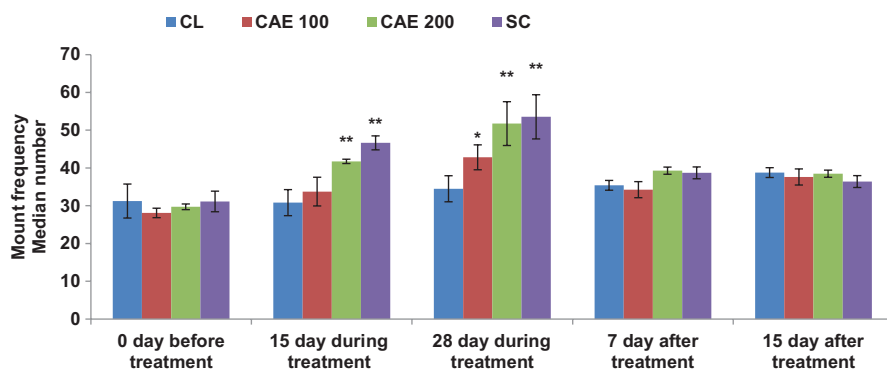


FIGURE 5 Effect of hydro-alcoholic extract of *Cassia auriculata* flower on mount frequency in male rats. Values are mean $\pm$ SD; $n = 6$; * $p < 0.05$, ** $p < 0.01$ vs. control

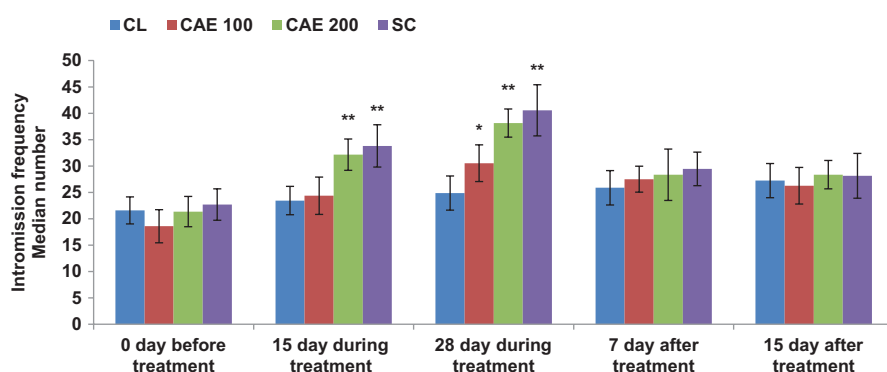


FIGURE 6 Effect of hydro-alcoholic extract of *Cassia auriculata* flower on intromission frequency in male rats. Values are mean $\pm$ SD; $n = 6$; * $p < 0.05$, ** $p < 0.01$ vs. control

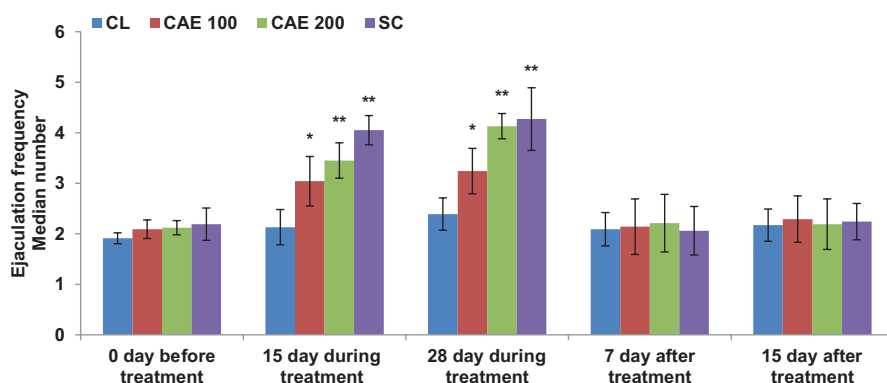


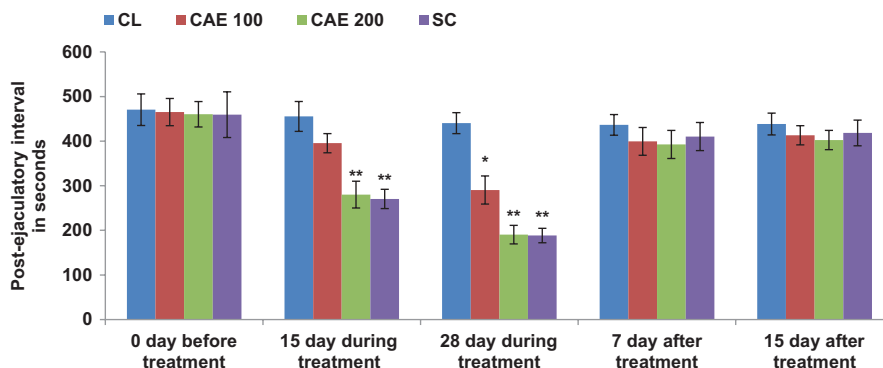
FIGURE 7 Effect of hydro-alcoholic extract of *Cassia auriculata* flower on ejaculation frequency in male rats. Values are mean $\pm$ SD; $n = 6$; * $p < 0.05$, ** $p < 0.01$ vs. control

traditional medicine for centuries without any known or recorded toxicity. Moreover, various parts of this plant extracted using different solvents were also found safe in the literature (Gupta et al., 2009; Pari, & Latha, 2002). However, the chemical constituents and the activities of this plant extracts are extremely varied with respect to the part used as well as the solvent employed for extraction (Boeing et al., 2014). Therefore, we have investigated the toxicity of the hydro-alcoholic extract of the flower part in the present work. OECD guideline was adopted to ascertain the toxicity of CAE using multiple doses including reasonably high doses of the drug (Babu, Gangudevi, & Subramaniam, 2003). In this connection, the test drug was subjected to an acute toxicity testing up to a high concentration of 2000 mg/kg, orally (ten times more than the aphrodisiac dose, evaluated in the present study). Even

at this dose, the extract did not produce any signs of toxicity or treatment-related adverse effects in the tests for aphrodisiac activity. This demonstrates that its short-term use for this purpose is apparently safe.

The investigations carried out with the CAE provide experimental evidence that the flowers of this plant are a potent stimulator of sexual behaviour. Improved sexual arousal, stimulation and penile erections were observed in the treated animals. In the first part of the functional study, the oral administration of CAE in male rats increased the penile erection index (PEI) in treated animals. This is indicative of the high degree of sexual stimulation. It was also observed that a significant change in mounting and intromission latency and frequency was observed at 15 and 28 days of the treatment. Here, we demonstrated that the oral administration of

FIGURE 8 Effect of hydro-alcoholic extract of *Cassia auriculata* flower on post-ejaculatory interval in male rats. Values are mean $\pm$ SD; $n = 6$; * $p < 0.05$, ** $p < 0.01$ vs. control



hydro-alcoholic extract of *C. auriculata* flower was able to improve sexual performance in male rats.

The post-ejaculatory interval is considered an index of potency, libido and the rate of recovery from exhaustion after the first series of mating (Tajuddin et al., 2004). A post-ejaculatory interval of more than 5,400 s indicates that the male is sexually exhausted and the intensity of sexual behaviour will be reduced in subsequent mating (Agmo, 1997). Therefore, the significantly decreased post-ejaculatory interval at 100 and 200 mg/kg body of CAE on day 28 may be attributed to enhanced potency and libido or less exhaustion in the first series of mating or both. In addition, the higher values of the computed male rat sexual behaviour parameters following treatment with the extract when compared with the control animals are indications of significant and sustained increase in sexual activity. However, the high penile erection index and all other sexual behaviour parameter were slowly got normalised in the treatment animals when they were evaluated after a lapse of 7 days and 15 days of withdrawal of treatment. The animals treated with sildenafil citrate also did not sustain the effect, and the penile erection index of this group was comparable to untreated control group animals. However, we have observed a slight but non-significant increment (Figures 1–8) in the sexual behaviour after withdrawal of treatment when compared with the control group. This may be due to the effect of experienced rats, since the sexual performance of the male rats is gradually improved with experience (Olivier et al., 2011).

Sildenafil citrate, a known phosphodiesterase-5 (PDE-5) inhibitor, has been used as a positive control based on the literature (Melnyk, & Marcone, 2011). During sexual arousal, nitric oxide (NO) is released from nerve terminals and endothelial cells in the corpus cavernosum (CC). The released NO activates guanylyl cyclase to produce cyclic GMP (cGMP) a potent vasodilator. The accumulation of cGMP leads to smooth-muscle relaxation in the CC and increased blood flow to the penis. PDE5 is an enzyme found primarily in the smooth muscle of the CC that selectively hydrolysis cGMP to 5'-GMP. However, PDE5 inhibitors block the cGMP hydrolysis, thus enhancing the effects of NO and leads to a prolonged erection. PDE5 inhibitors show that the lack of direct effect on CC smooth-muscle relaxation. Therefore, after administration, adequate sexual stimulation is necessary for an erection to occur (Huang, & Lie, 2013). On the other hand, antioxidants are believed to ameliorate

TABLE 1 Total phenolic and flavonoid content of CAE

Total phenolic content ^a	141.07 $\pm$ 6.71
Total flavonoid content ^b	62.24 $\pm$ 3.27

Notes. Values are mean $\pm$ SD of three replicates.

^amg gallic acid equivalent (GAE)/g dry weight. ^bmg rutin equivalent/g dry weight.

the erectile response of patients with erectile dysfunction (ED). Vicari, La Vignera, Condorelli, and Calogero (2010) reported that administration of endothelial antioxidant compounds to patients with arterial ED had improved the success rate of sildenafil treatment. These data suggest that, in such patients, a combined treatment may be considered to increase bioavailable NO and to neutralise radical oxygen species, which in turn inactivate NO. Phenolic compounds have redox properties, which allow them to act as antioxidants (Soobrattee, Neergehen, Luximon-Ramma, Aruoma, & Bahorun, 2005). Flavonoids, including flavones, flavanols and condensed tannins, are plant secondary metabolites, well documented as antioxidants both *in vitro* and *in vivo*, due to the presence of three OH groups especially the OH at 3rd position (Shimoi, Masuda, Shen, Furugori, & Kinze, 1996). Various flavonoids isolated from different plant species have shown inhibitory effect on several isoforms of PDE along with antioxidant properties. Quercetin, a known antioxidant present in *Allium cepa*, has been reported to possess PDE5 inhibitory effect (Lines, & Ono, 2006). Sánchez-Mendoza et al. (2007) reported that flavonoids in *Gnaphalium liebmanni* may have PDE inhibitory effect. Anthocyanin from *Vitis vinifera* also showed the inhibitory activity on PDE5 (Dell'Agli et al., 2005).

The recent literature reported that *C. auriculata* flowers are rich in tannins and flavonoids (Juan-Badaturuge, Habtemariam, & Thomas, 2011; Purushotham, Annegowda, Sathish, Ramesh, & Mansor, 2014). Hence, the quantitative estimation of phenolic and flavonoid contents was carried out to ascertain the possible correlation of observed activity. In the present investigation, the copulatory behaviours of both CAE and sildenafil citrate-treated animals were appearing to be similar when compared with the control group. These findings suggest that CAE may act the similar mode of action as of sildenafil citrate, either through phosphodiesterase-5 (PDE-5) inhibitory pathway or the enhanced erectile response through

antioxidant property due to the presence of phenolic compounds or both in synergism. However, further investigations are in progress to ascertain the mechanism of action of CAE along with studying the hormone levels and histological parameters. Activity-guided isolation of PDE-5 inhibitory principle is also going on in our laboratory. Further, the effect of the drug on female sexual behaviour and fertility remains to be studied.

5 | CONCLUSION

This is the first report, which provides the scientific data to confirm the *C. auriculata*, possesses the aphrodisiac effects in Wister albino male rats. The increase in copulatory behaviours induced by the plant extract may be due to the presence of phenolic compounds in the extract. Our previous report on the antioxidant activity of this plant also supports that it could be further developed as the functional food or health product for men especially for who are risks for sexual dysfunction. However, further investigation about the active ingredients exhibiting these effects and the precise underlying mechanism of the extract is still required.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interests.

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REFERENCES

- Acute Oral Toxicity Test Method (2001). *OECD Guidelines for testing of Chemicals*. Paris: Organization for Economic Cooperation and Development.
- Agmo, A. (1997). Male rat sexual behaviour. *Brain Research Protocols*, 1, 203–209.
- Agmo, A. (2003). Unconditioned sexual incentive motivation in the male Norway rat (*Rattus norvegicus*). *Journal of Comparative Psychology*, 117, 3–14. <https://doi.org/10.1037/0735-7036.117.1.3>
- Almeida, S. A., Kempinas, W. G., & Lamano-Carvalho, T. L. (2000). Sexual behavior and fertility of male rats submitted to prolonged immobilization-induced stress. *Brazilian Journal of Medical and Biological Research*, 33, 1105–1109. <https://doi.org/10.1590/S0100-879X2000000900019>
- Annie, S., Rajagopal, P. L., & Malini, S. (2005). Effect of Linn root extract on cisplatin and gentamicin-induced renal injury. *Phytomedicine*, 12, 555–560.
- Arthanari, S., Vanitha, J., Krishnaswami, V., Renukadevi, P., Deivasigamani, K., & De Clercq, E. (2013). Antiviral and cytotoxic screening of methanolic extract of flowers in HeLa, Vero, CRFK and HEL cell lines. *Drug Invention Today*, 5, 28–31.
- Babu, V., Gangudevi, T., & Subramaniam, A. (2003). Antidiabetic activity of ethanol extract of *Cassia kleinii* leaf in streptozotocin - induced diabetic rats and isolation of an active fraction and toxicity evaluation of the extract. *Indian Journal of Pharmacology*, 35, 290–296.
- Boeing, J. S., Barizão, É. O., Silva, B. C., Montanher, P. F., de Cinque Almeida, V., & Visentainer, J. V. (2014). Evaluation of solvent effect on the extraction of phenolic compounds and antioxidant capacities from the berries: Application of principal component analysis. *Chemistry Central Journal*, 8, 48. <https://doi.org/10.1186/s13065-014-0048-1>
- Brahmachari, H. D., & Augusti, K. T. (1961). Hypoglycaemic Agents from Indian Indigenous Plants. *Journal of Pharmacy and Pharmacology*, 13, 381–382. <https://doi.org/10.1111/j.2042-7158.1961.tb11839.x>
- Butcher, L. L., Butcher, S. G., & Larsson, K. (1969). Effects of apomorphine, (+)- amphetamine, and nialamide on tetra-benzazine-induced suppression of sexual behavior in the male rat. *European Journal of Pharmacology*, 7, 283–288. [https://doi.org/10.1016/0014-2999\(69\)90093-4](https://doi.org/10.1016/0014-2999(69)90093-4)
- Chang, C., Yang, M., Wen, H., & Chern, J. (2002). Estimation of total flavonoid content in propolis by two complementary colorimetric methods. *Journal of Food and Drug Analysis*, 10, 178–182.
- Chauhan, K. N., Patel, M. B., Valera, H. R., Patil, S. D., & Surana, S. J. (2009). Hepatoprotective activity of flowers of *Cassia auriculata* R.Br. against paracetamol induced liver injury. *Journal of Natural Remedies*, 9, 85–89.
- Dell'Agli, M., Galli, G. V., Vrhovsek, U., Mattivi, F., & Bosio, E. (2005). In vitro inhibition of human cGMP-specific phosphodiesterase-5 by polyphenols from red grapes. *Journal of Agricultural and Food Chemistry*, 53, 1960–1965. <https://doi.org/10.1021/jf048497+>
- Dewsbury, D. A., & Davis, H. N. (1970). Effects of reserpine on the copulatory behavior of male rats. *Physiology & Behavior*, 5, 1331–1333. [https://doi.org/10.1016/0031-9384\(70\)90050-8](https://doi.org/10.1016/0031-9384(70)90050-8)
- Gupta, S., Sharma, S. B., Bansal, S. K., & Prabhu, K. M. (2009). Antihyperglycemic and hypolipidemic activity of aqueous extract of *Cassia auriculata* L. leaves in experimental diabetes. *Journal of Ethnopharmacology*, 123, 499–503. <https://doi.org/10.1016/j.jep.2009.02.019>
- Huang, S. A., & Lie, J. D. (2013). Phosphodiesterase-5 (PDE5) inhibitors in the management of erectile dysfunction. *A Peer-Reviewed Journal for Formulary Management*, 38, 407–419.
- Islam, M. W., Tariq, M., Ageel, A. M., Al-said, M. S., & Al-Yhya, A. M. (1991). Effects of *Salvia haematodes* on sexual behaviour of male rats. *Journal of Ethnopharmacology*, 33, 67–72.
- Jackson, G. (1998). Sildenafil (Viagra) "reaches the parts other drugs cannot reach. *International Journal of Clinical Practice*, 52, 359.
- Jayaweera, D. M. A. (1981). *Medicinal plants (indigenous and exotic) used in Ceylon, Part 3*. Colombo: The National Science Foundation.
- Joshi, S. G. (2000). *Cesalpiniaceae. Text book of medicinal plants* (p. 119). New Delhi, India: Oxford and IBH Publishing.
- Juan-Badaturge, M., Habtemariam, S., & Thomas, M. J. K. (2011). Antioxidant compounds from a South Asian beverage and medicinal plant, *Cassia auriculata*. *Food Chemistry*, 125, 221–225. <https://doi.org/10.1016/j.foodchem.2010.08.065>
- Karras, D. J., Farrell, S. E., Harrigan, R. A., Henretig, F. M., & Gealt, L. (1996). Poisoning from "Spanish fly" (cantharidin). *The American Journal of Emergency Medicine*, 14(5), 478–483. [https://doi.org/10.1016/S0735-6757\(96\)90158-8](https://doi.org/10.1016/S0735-6757(96)90158-8)
- Kaur, C., & Kapoor, H. C. (2002). Anti-oxidant activity and total phenolic content of some Asian vegetables. *International Journal of Food Science & Technology*, 37, 153–161. <https://doi.org/10.1046/j.1365-2621.2002.00552.x>

- Kenmogne, H., Koloko, B., Hambe, C., Domkam, J., Ngaha-Njila, M., Bend, E., ... Dimo, T. (2016). Effects of aqueous root extract of *Carpolobia alba* G. Don on sexual behaviour in adult male rats. *Andrologia*, 48, 908–914.
- Kirthkar, K. R., & Basu, B. D. (1965). *Medicinal plants of India*, Vol. I (pp. 518–520). New Delhi: Published by CSIR.
- Kumar, R. S., Ponmozhi, M., Viswanathan, P., & Nalini, N. (2003). Effect of *Cassia auriculata* leaf extract on lipids in rats with alcoholic liver injury. *Asia Pacific Journal of Clinical Nutrition*, 11, 157–163.
- Kumaran, A., & Karunakaran, R. J. (2007). Antioxidant activity of *Cassia auriculata* flowers. *Fitoterapia*, 78, 46–47. <https://doi.org/10.1016/j.fitote.2006.09.031>
- Lines, T. C., & Ono, M. (2006). FRS 1000, an extract of red onion peel, strongly inhibits phosphodiesterase 5A (PDE 5A). *Phytomedicine*, 13, 236–239. <https://doi.org/10.1016/j.phymed.2004.12.001>
- Malviya, N., Malviya, S., Jain, S., & Vyas, S. (2016). A review of the potential of medicinal plants in the management and treatment of male sexual dysfunction. *Andrologia*, 48, 880–893.
- Manimegalai, S., & Venkatalakshmi, P. (2012). Cardioprotective effect of *Cassia auriculata* Linn., petal extract on isoproterenol induced myocardial infarction in albino rats. *International Journal of Pharmaceutical Sciences and Research*, 3, 848–852.
- Melnik, J. P., & Marcone, M. F. (2011). Aphrodisiacs from plant and animal sources-A review of current scientific literature. *Food Research International*, 44, 840–850. <https://doi.org/10.1016/j.foodres.2011.02.043>
- Olivier, B., Chan, J. S., Snoeren, E. M., Olivier, J. D., Veening, J. G., Vinkers, C. H., ... Oosting, R. S. (2011). Differences in sexual behaviour in male and female rodents: Role of serotonin. *Biological Basis of Sex Differences in Psychopharmacology*, 8, 15–36.
- Pari, L., & Latha, M. (2002). Effect of *Cassia auriculata* flowers on blood sugar levels, serum and tissue, lipids in streptozotocin diabetic rats. *Singapore Medical Journal*, 43, 617–621.
- Prasanna, R., Harish, C. C., Pichai, R., Sakthisekaran, D., & Gunasekaran, P. (2009). Anti-cancer effect of *Cassia auriculata* leaf extract in vitro through cell cycle arrest and induction of apoptosis in human breast and larynx cancer cell lines. *Cell Biology International*, 33, 127–134. <https://doi.org/10.1016/j.cellbi.2008.10.006>
- Purushotham, K. N., Annegowda, H. V., Sathish, N. K., Ramesh, B., & Mansor, S. M. (2014). Evaluation of phenolic content and antioxidant potency in various parts of *Cassia auriculata* L.: A traditionally valued plant. *Pakistan Journal of Biological Sciences*, 17, 41–48.
- Saad, M. A., Eid, N. I., Abd El-Latif, H. A., & Sayed, H. M. (2013). Potential effects of yohimbine and sildenafil on erectile dysfunction in rats. *European Journal of Pharmacology*, 700, 127–133. <https://doi.org/10.1016/j.ejphar.2012.12.020>
- Sabu, C., & Subburaju, T. (2002). Effect of *Cassia auriculata* Linn. on serum glucose level, glucose utilization by isolated rat hemidiaphragm. *Journal of Ethnopharmacology*, 80, 203–206. [https://doi.org/10.1016/S0378-8741\(02\)00026-0](https://doi.org/10.1016/S0378-8741(02)00026-0)
- Sánchez-Mendoza, M. E., Torres, G., Arrieta, J., Aguilar, A., Castillo-Henkel, C., & Navarrete, A. (2007). Mechanisms of relaxant action of a crude hexane extract of *Gnaphalium liebmanni* in guinea pig tracheal smooth muscle. *Journal of Ethnopharmacology*, 111, 142–147.
- Sandroni, P. (2001). Aphrodisiacs past and present: A historical review. *Clinical Autonomic Research*, 11, 303–307. <https://doi.org/10.1007/BF02332975>
- Sharma, V., Thakur, M., & Dixit, V. K. (2012). A comparative study of ethanolic extracts of *Pedaliu murex* Linn. fruits and sildenafil citrate on sexual behaviors and serum testosterone level in male rats during and after treatment. *Journal of Ethnopharmacology*, 143, 201–206. <https://doi.org/10.1016/j.jep.2012.06.024>
- Shimoi, K., Masuda, S., Shen, B., Furugori, M., & Kinze, N. (1996). Radioprotective effects of antioxidative plant flavonoids in mice. *Mutation Research*, 350, 153–161. [https://doi.org/10.1016/0027-5107\(95\)00116-6](https://doi.org/10.1016/0027-5107(95)00116-6)
- Singh, G., & Mukherjee, T. (1988). Herbal aphrodisiac: A review. *Indian Drugs*, 35, 175–182.
- Siva, R., & Krishnamurthy, K. V. (2005). Isozyme diversity in *Cassia auriculata* L. *African Journal of Biotechnology*, 4, 772–775.
- Soobrattee, M. A., Neergheen, V. S., Luximon-Ramma, A., Aruoma, O. I., & Bahorun, O. T. (2005). Phenolics as potential antioxidant therapeutic agents: Mechanism and actions. *Mutation Research*, 579, 200–213. <https://doi.org/10.1016/j.mrfmmm.2005.03.023>
- Surana, S., Gokhale, B., Jadhav, R. B., Sawant, R. L., & Wadekar, J. B. (2008). Antihyperglycemic activity of various fractions of *Cassia auriculata* linn. in alloxan diabetic rats. *Indian Journal of Pharmaceutical Sciences*, 70, 227–229.
- Taha, S. A., Islam, M. W., & Ageel, A. M. (1995). Effect of ambrein, a major constituent of ambergris, on masculine sexual behavior in rats. *Archives Internationales De Pharmacodynamie Et De Therapie*, 329, 283–294.
- Tajuddin, A., Ahmad, S., Latif, A., & Qasmi, I. A. (2004). Effect of 50% ethanolic extract of *Syzygium aromaticum* (L.) Merr. & Perry. (clove) on sexual behaviour of normal male rats. *BMC Complementary and Alternative Medicine*, 4, 17–24.
- Thakur, M., Thompson, D., Connellan, P., Deseo, M. A., Morris, C., & Dixit, V. K. (2011). Improvement of penile erection, sperm count and seminal fructose levels in vivo and nitric oxide release in vitro by ayurvedic herbs. *Andrologia*, 43, 273–277. <https://doi.org/10.1111/j.1439-0272.2010.01068.x>
- Vedavathy, S., & Rao, K. N. (1991). Antipyretic activity of six indigenous medicinal plants of Tirumala hills, Andhra Pradesh, India. *Journal of Ethnopharmacology*, 33, 193–196. [https://doi.org/10.1016/0378-8741\(91\)90178-G](https://doi.org/10.1016/0378-8741(91)90178-G)
- Vicari, E., La Vignera, S., Condorelli, R., & Calogero, A. E. (2010). Endothelial antioxidant administration ameliorates the erectile response to PDE5 regardless of the extension of the atherosclerotic process. *The Journal of Sexual Medicine*, 7, 1247–1253. <https://doi.org/10.1111/j.1743-6109.2009.01420.x>

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